Simple and Compound Interest

My Business Math reviewer! Interest is the price you pay for borrowing money, or the reward you earn for letting someone else (like a bank or cooperative) use your money.

Key terms

- Principal (P): the original amount borrowed, deposited, or invested.
- Rate (r): the percentage charged or earned per year, written as a decimal when computing. So 6% becomes 0.06.
- Time (t): how long the money is borrowed or invested, always expressed in years when the rate is yearly.
- Interest (I): the extra money earned or paid on top of the principal.
- Maturity value or future value (F): the total amount at the end, which is principal plus interest.
- Lender or investor: the one who gives the money. Borrower or debtor: the one who receives it and must pay back.

Simple interest

In simple interest, interest is computed only on the original principal for the whole term. The interest earned each year stays the same because the base never changes.

The formula is I = P x r x t, often shortened to I = Prt.

The maturity value is F = P + I, which can also be written F = P(1 + rt).

Example 1. Ana's family borrows ₱50,000 from a lending cooperative at 6% simple interest for 3 years.

- I = 50,000 x 0.06 x 3 = ₱9,000
- F = 50,000 + 9,000 = ₱59,000

Example 2. Time given in months. Carlo deposits ₱20,000 at 5% simple interest for 9 months. Convert months to years first: 9 / 12 = 0.75 year.

- I = 20,000 x 0.05 x 0.75 = ₱750
- F = ₱20,750

When time is in days, there are two conventions. Exact interest divides the number of days by 365. Ordinary interest divides by 360, which makes the computed interest slightly bigger. Always check which one the problem asks for.

You can also rearrange the formula to find a missing value: P = I / (rt), r = I / (Pt), and t = I / (Pr).

Compound interest

In compound interest, the interest earned is added to the principal at the end of each period, and the next interest is computed on that bigger amount. In short, you earn interest on your interest. This is why compound interest grows faster than simple interest over the same time and nominal rate.

The formula is F = P(1 + r/m)^(mt)

- m is the number of compounding periods in one year.
- r/m is the interest rate per period.

- mt is the total number of compounding periods in the whole term.
- Compound interest is then I = F - P.

Common compounding frequencies:

- Annually: m = 1
- Semi-annually: m = 2
- Quarterly: m = 4
- Monthly: m = 12
- Daily: m = 365 in most textbook problems

Example 3. Same ₱50,000 at 6% for 3 years, but compounded quarterly.

- r/m = 0.06 / 4 = 0.015
- mt = 4 x 3 = 12 periods
- F = 50,000(1.015)^12 = about ₱59,780.91
- Compound interest = 59,780.91 - 50,000 = ₱9,780.91

Compare this with simple interest in Example 1, which gave only ₱9,000. Compounded annually instead, the same loan reaches 50,000(1.06)^3 = ₱59,550.80. So the more often interest is compounded, the bigger the final amount.

Present value

Sometimes you know the target amount and need to find how much to put in today. Just solve the compound formula for P:

P = F / (1 + r/m)^(mt)

Example 4. Bea wants ₱100,000 for her college fees in 2 years. Her bank pays 4% compounded semi-annually. Then r/m = 0.02 and mt = 4.

- P = 100,000 / (1.02)^4 = about ₱92,384.54

She needs to deposit roughly ₱92,385 today.

Nominal rate versus effective rate

The nominal rate is the stated yearly rate, like "6% compounded quarterly." The effective rate is the rate that, if compounded only once a year, would give the same yearly growth. It shows the true annual cost or earning.

The formula is: effective rate = (1 + r/m)^m - 1

- 6% compounded quarterly: (1.015)^4 - 1 = 0.0614, or about 6.14% effective.
- 12% compounded monthly: (1.01)^12 - 1 = 0.1268, or about 12.68% effective.

The effective rate is always higher than the nominal rate whenever compounding happens more than once a year. When m = 1, the two rates are equal. When comparing banks, pick the higher effective rate for saving and the lower one for borrowing.

Quick check

- Q: What is the simple interest on ₱10,000 at 8% for 2 years? A: I = 10,000 x 0.08 x 2 = ₱1,600.
- Q: In F = P(1 + r/m)^(mt), what does m stand for? A: The number of compounding periods per year.

- Q: What is the value of m for quarterly compounding? A: 4.
- Q: Which is bigger for the same principal, rate, and time: simple or compound interest? A: Compound interest, as long as there is more than one period, because interest also earns interest.
- Q: What is the effective rate of 12% compounded monthly? A: About 12.68%.